

Ye Marn Aung

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Professional Summary

Computer Science senior with a proven track record delivering full-stack, AI-enhanced products from concept through production. Expert in **TypeScript**, **JavaScript/Node.js**, and **Python**-based microservices (FastAPI, Flask). Strong experience designing **RESTful APIs**, building CI/CD pipelines with **GitHub Actions**, and deploying scalable services on **AWS EC2** (Docker) while leveraging cloud storage solutions. Demonstrated leadership of remote, Agile teams, driving high-impact outcomes and top-rated project deliveries. Seeking a role that blends AI/ML expertise with robust software engineering to create user-centric, cloud-native solutions.

Skills

Languages & Frameworks	Cloud & DevOps	Databases & Storage	AI/ML & Data	Collaboration & Tools
TypeScript, JavaScript, Node.js, React, Next.js	AWS EC2, S3 (static asset hosting), Docker, GitHub Actions, CI/CD pipelines	MySQL, PostgreSQL, MongoDB, Pinecone (vector DB)	TensorFlow, Keras, PyTorch, HuggingFace, Scikit-Learn, OpenCV	Git/GitHub, GitHub Issues/Projects, Slack, Zoom, Agile Scrum, Figma, Google Gemini

Experience

Team Lead / Database Engineer – GaitorGate (AI-Powered Search Engine)

San Francisco State University – Remote | Jan 2024 – Present

- Led a 6-member multidisciplinary team using Agile sprints, daily stand-ups via **Slack** and weekly video syncs, delivering the highest-rated product among 12 competing groups.
- Designed and implemented **RESTful Flask APIs** for authentication, NLP-enhanced search, user ratings, and a Google Gemini-powered chatbot; documented endpoints with OpenAPI specifications.
- Architected a microservice layout separating the API layer (Gunicorn) from the static front-end, containerized with **Docker**, and deployed on **AWS EC2** with **GitHub Actions** automating tests, builds, and production roll-out.
- Managed MySQL schema design, indexing, and migration scripts; ensured data integrity for >200,000 search queries.
- Coordinated version control and code reviews through **GitHub**, enforcing branch protection and pull-request workflows.

Full-Stack Developer – MovieCenter (RAG-Based LangChain Recommender)

Personal Project – Remote | Sep 2023 – Dec 2023

- Built a **FastAPI** backend exposing 12+ endpoints (search, recommendation, user preferences) following REST conventions; generated automatic API docs with Swagger UI.
- Integrated **LangChain**, Pinecone vector store, and MongoDB to serve context-rich movie recommendations for ~10 K titles; improved semantic similarity recall by **30%** using HuggingFace SentenceTransformers.
- Developed a **Next.js** front-end with **TypeScript** components, leveraging React hooks for real-time UI updates.
- Implemented CI/CD using **GitHub Actions** to lint, run unit tests (pytest), build Docker images, and push to AWS EC2 for zero-downtime deployments.
- Utilized **AWS S3** to host static assets (movie poster thumbnails) and served them via CloudFront, reducing latency by 22%.

Developer – Board2Board (Chess Utility AI)

Personal Project – Remote | Mar 2023 – Aug 2023

- Designed a computer-vision pipeline with **OpenCV** to segment chessboard images into 64 squares, followed by a **ResNet50** model fine-tuned on 2000+ labeled images (94 % accuracy across 13 classes).
- Added a linear-regression-based adaptive thresholding module to handle variable lighting,

boosting robustness across diverse board captures.

- Produced reproducible Jupyter notebooks and packaged the model with **Joblib** for easy inference in downstream applications.

Projects

- **AI-Driven Search Engine (GaitorGate)** – Full-stack solution with NLP search, AI chatbot, and secure user management; deployed on AWS EC2 with automated CI/CD.
 - **Movie Recommendation System (MovieCenter)** – RAG pipeline leveraging LangChain, vector DB, and Gemini; TypeScript front-end with FastAPI microservice backend, containerized and cloud-hosted.
 - **Chessboard Recognition (Board2Board)** – End-to-end computer vision and deep-learning system for real-time OTB chess piece detection.
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Education

B.S. Computer Science (Minor: Mathematics)

San Francisco State University – San Francisco, CA | Expected Dec 2026

Certifications

Pending – Pursuing AWS Certified Solutions Architect – Associate (expected 2025)

Technical Tools

- **Version Control:** Git, GitHub (Actions, Projects, Issues)
- **Containerization:** Docker, Docker Compose
- **Cloud Services:** AWS EC2, S3, IAM, CloudFront

- **CI/CD:** GitHub Actions (lint, test, build, deploy)
 - **Project Management:** Agile Scrum, Jira (basic), Trello, Slack, Zoom
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